

MIRPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

NAME:

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ROLL NO:

FA23-BSE-003

SUBJECT:

SQE

DATE:

05 MAY 2026

INSTRUCTOR:

ENGR. ASIM JAVAID

REPORT TYPE:

LAB TASK

DEPARTMENT:

SOFTWARE ENGINEERING

Test Plan

SECTION 1 — TEST PLAN IDENTIFIER

Test Plan ID	LMS-TP-001
Version	1.0
Date	05 May 2026
Project	University Library Management System
Prepared By	Abdur Rehman- SQE-lab-student
Reviewed By	Engr. Asim Javaid
Status	Draft- Pending Review

SECTION 2 — INTRODUCTION

The University Library Management System (LMS) is a web-based application developed for a university to manage book loans, returns, student accounts, and library staff operations. The system serves two user roles: Students and Library Staff, interacting through a browser-based interface backed by a PostgreSQL database and Node.js server.

SECTION 3 — TEST ITEMS

Item	Version / Build	Description
LMS Frontend (React.js)	v1.0.0 (Build #42)	Browser-rendered React application accessible through web browser
LMS Backend (Node.js/Express)	v1.0.0	REST API server handling business logic and database communication
LMS Database (PostgreSQL)	v14.2	Relational database storing books, students, staff, and loan records
Authentication Service	v1.0.0	JWT-based authentication service managing sessions and password reset
Test Data Set	LMS-TESTDATA-001	Pre-populated database containing 200 books, 50 students, and 5 staff accounts

Chrome Browser	v120+	Primary supported browser for testing
JIRA Test Management Project	sqe-lab-abdurrehman	Test management environment used for issue tracking and documentation

SECTION 4 — FEATURES TO BE TESTED

Feature	Requirement ID	Priority
User login using valid credentials	REQ-01	High
Rejection of invalid login attempts	REQ-01	High
Account lockout after 3 failed login attempts	REQ-02	High
Account lockout notification display	REQ-02	Medium
Password reset request functionality	REQ-03	Medium
Password reset email delivery within 2 minutes	REQ-03	Medium
Session automatic expiry after 30 minutes of inactivity	REQ-04	High
Session timer reset after user activity	REQ-04	Medium
Search books by title	REQ-05	High
Search books by author	REQ-05	High
Search books by ISBN	REQ-05	High
Filter search results by genre	REQ-06	Medium
Filter search results by availability status	REQ-06	Medium
Search response time under 2 seconds	REQ-07	High
Pagination for result sets exceeding 20 items	REQ-08	Low
Validation of search result accuracy	REQ-05	High
Verification of session security behavior	REQ-04	High

SECTION 5 — FEATURES NOT TO BE TESTED

Feature NOT Tested	Justification
Loan Management Module (REQ-09 to REQ-12)	Out of scope for Week 3 testing cycle. Planned for future test plan LMS-TP-002
Notification Module (REQ-13 to REQ-16)	Notification infrastructure and SMS gateway are not configured in current environment
Load and Stress Testing above 100 concurrent users	Performance infrastructure and load testing tools are unavailable during this lab session
Cross-browser compatibility testing	Current testing cycle officially supports Chrome browser only
Mobile responsiveness testing	LMS v1.0 is desktop-focused and mobile optimization is planned for future release
API penetration testing	Security penetration testing is scheduled for future cybersecurity lab sessions
Database backup and recovery testing	Managed by infrastructure and DevOps team
Accessibility compliance testing	Accessibility testing tools are not configured for this test cycle

Automation testing framework validation	Selenium and automated frameworks will be introduced in later labs
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SECTION 6 — TEST APPROACH

6.1 Testing Level

The LMS application will undergo System Testing. Testing will validate the complete integrated application including frontend interface, backend services, authentication mechanisms, and database interactions.

6.2 Testing Type

The primary testing method will be Functional Black-Box Testing. Additional performance testing will be conducted for requirements related to response time and email delivery.

6.3 Testing Techniques

The following software testing techniques will be used:

- Black-Box Testing
- Equivalence Partitioning
- Boundary Value Analysis
- Positive and Negative Testing
- Functional Validation
- Basic Performance Testing
- Session Management Validation

6.4 Execution Method

All tests during this cycle will be executed manually by the testing team using predefined test cases documented in JIRA.

6.5 Test Priority Order

Testing will be conducted in the following order:

1. High Priority Requirements
2. Medium Priority Requirements
3. Low Priority Requirements

Critical authentication and session management features will be tested before search functionality.

6.6 Defect Management

All identified defects will be logged into JIRA using the standard defect reporting format. Each defect will include:

- Defect ID
- Severity
- Priority
- Steps to reproduce
- Expected result
- Actual result
- Screenshots
- Environment details

6.7 Re-testing

All defects resolved by developers will be re-tested within 24 hours of fix deployment.

6.8 Regression Testing

After every major defect fix, smoke testing will be executed to verify that core application functionality continues to operate correctly.

6.9 Test Data Management

Testing will use a dedicated test dataset containing:

- 200 books
- 50 student accounts
- 5 staff accounts
- Sample ISBN records
- Genre-specific datasets
- Availability status records

The database will be restored using snapshots between testing cycles to maintain data consistency.

SECTION 7 — PASS / FAIL CRITERIA

7.1 Test Case Pass Criteria

A test case shall be considered PASS when:

- Actual result exactly matches expected result
- No unexpected behavior occurs
- No system crash or data corruption is observed
- UI elements function correctly

- Validation messages appear correctly

7.2 Test Case Fail Criteria

A test case shall be considered FAIL when:

- Actual result differs from expected result
- Incorrect data is displayed
- System crashes or freezes
- Validation behavior is incorrect
- Performance exceeds acceptable limits
- Security behavior does not function correctly

7.3 Test Cycle Exit Criteria

Testing will be considered complete when:

- 100% of High Priority test cases are executed
- 100% of Medium Priority test cases are executed
- No Critical defects remain open
- No High severity defects remain unresolved
- RTM coverage reaches 100%
- Final test report is reviewed and approved

7.4 Defect Severity Levels

Critical Severity

- System crash
- Complete login failure
- Database corruption
- Authentication bypass vulnerability

High Severity

- Session timeout not functioning
- Search functionality unavailable
- Incorrect account lockout behavior

Medium Severity

- Incorrect validation message
- Pagination alignment issues
- Minor workflow interruptions

Low Severity

- UI formatting issues

- Typographical errors
- Cosmetic display problems

SECTION 8 — SUSPENSION CRITERIA & RESUMPTION REQUIREMENTS

8.1 Suspension Criteria

Testing activities shall be suspended if any of the following conditions occur:

- LMS server becomes unavailable
- Database connection fails
- Critical defect prevents further testing
- Authentication module becomes inaccessible
- Test environment crashes repeatedly
- Test data becomes corrupted
- Network connectivity issues prevent access to the application

8.2 Resumption Requirements

Testing activities may resume only after:

- Failed systems are restored
- Critical defects are resolved
- Environment stability is verified
- Test database is restored successfully
- Smoke testing confirms system availability
- Approval is provided by the project supervisor or instructor

SECTION 9 — TEST DELIVERABLES

The following deliverables will be produced during this testing cycle:

Deliverable	Description
Requirement Traceability Matrix (RTM)	Mapping between requirements and test cases
IEEE 829 Test Plan	Official test planning document
Test Cases	Detailed manual test case specifications
Defect Reports	Documentation of identified defects
Test Execution Reports	Results of executed test cases
JIRA Project Board	Configured project management environment
Coverage Summary Report	Requirement coverage statistics

Screenshots	Evidence of testing activities and JIRA configuration
Sprint Configuration	Active sprint setup documentation
Final Submission Package	Complete set of lab deliverables

SECTION 10 — ENVIRONMENTAL NEEDS

10.1 Hardware Requirements

- Intel Core i5 or higher processor
- Minimum 8 GB RAM
- Stable internet connection
- Minimum 10 GB free disk space

10.2 Software Requirements

Software	Version
Windows Operating System	Windows 10/11
Google Chrome	v120+
Microsoft Excel or Google Sheets	Latest
Microsoft Word or Google Docs	Latest
JIRA Software	Cloud Version

10.3 Server Environment

Component	Environment
Frontend Hosting	AWS EC2
Backend Server	Node.js Express Server
Database	PostgreSQL v14.2
Authentication	JWT-based service

10.4 Test Data Requirements

- 200 books in database
- Student test accounts
- Staff test accounts
- Sample ISBN records
- Mixed availability statuses
- Genre-based records

10.5 Browser Configuration

Testing will be conducted using:

- Google Chrome browser only

- Cookies enabled
- JavaScript enabled
- Stable internet connection

SECTION 11 — RESPONSIBILITIES

Role	Responsibility
Test Engineer / Student	Design and execute test cases
Developer	Fix identified defects
Instructor	Review deliverables and provide feedback
Database Administrator	Maintain test database integrity
Project Manager	Monitor project progress and schedules
QA Team	Verify defect fixes and perform regression testing

11.1 Student Responsibilities

The student tester is responsible for:

- Preparing RTM documentation
- Writing and executing test cases
- Logging defects into JIRA
- Maintaining evidence of testing activities
- Exporting deliverables in required format
- Meeting submission deadlines

11.2 Instructor Responsibilities

The instructor is responsible for:

- Reviewing deliverables
- Providing technical guidance
- Verifying completeness of submissions
- Assessing lab performance

SECTION 12 — TEST SCHEDULE

Activity	Start Date	End Date	Duration
Requirements Analysis	Day 1	Day 1	2 Hours
JIRA Project Configuration	Day 1	Day 1	35 Minutes
RTM Development	Day 1	Day 1	45 Minutes
Test Plan Documentation	Day 1	Day 1	40 Minutes
Test Case Design	Day 1	Day 1	35 Minutes
Defect Logging Setup	Day 1	Day 1	15 Minutes
Final Review	Day 1	Day 1	20 Minutes
Submission Preparation	Day 1	Day 1	15 Minutes

12.1 Milestones

Milestone	Expected Completion
JIRA Configuration Completed	End of Experiment A
RTM Completed	End of Experiment B
IEEE 829 Test Plan Completed	End of Experiment C
Test Cases Created	End of Experiment D
Final Submission Ready	End of Lab Session

